



Stony Brook
University

Fishing in Deeper Waters: A Comprehensive MEV Analysis of DAG-Based Consensus Protocols

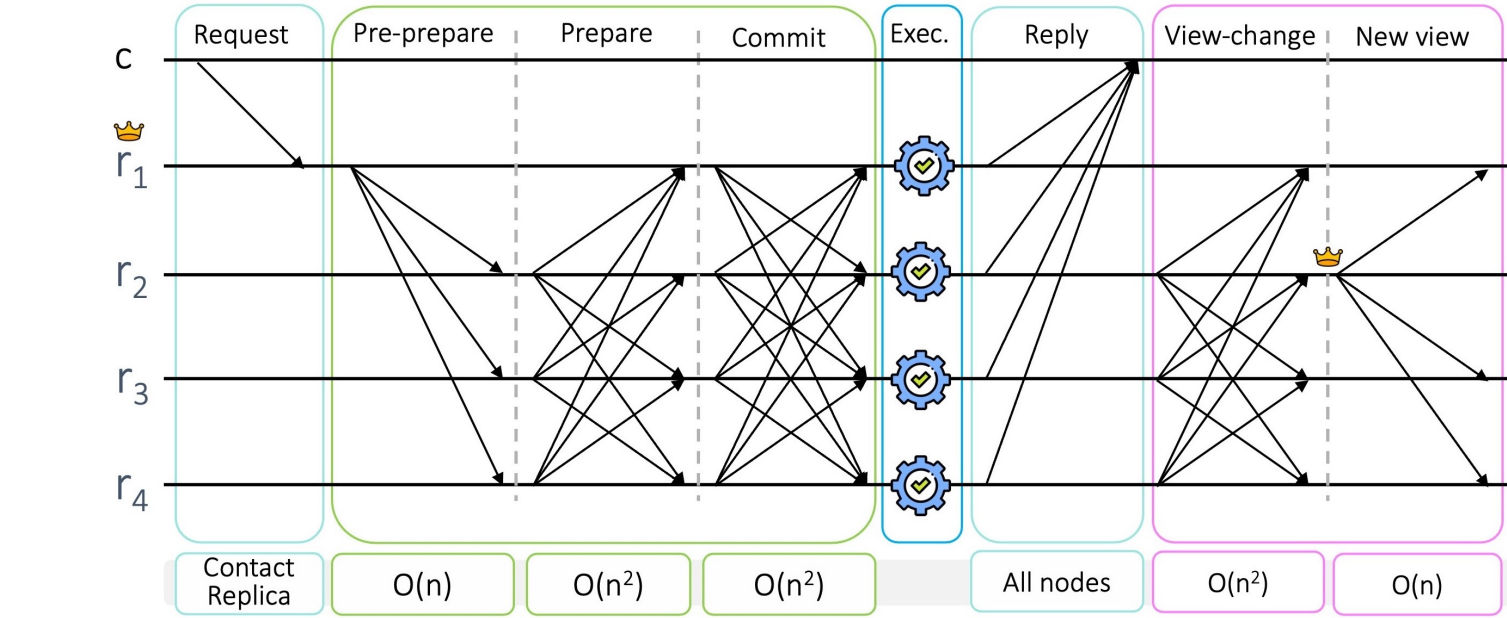
Iliya Mirzaei, Mohammad Javad Amiri
Department of Computer Science, Stony Brook University, NY, USA

Abstract

Today's large-scale distributed systems operate in untrustworthy environments where Byzantine Fault-Tolerant (BFT) consensus protocols ensure non-faulty nodes execute client requests in the same order. While traditional protocols often rely on static configuration parameters, this research investigates the security implications of shifting to Directed Acyclic Graph (DAG) based BFT protocols. We present a comprehensive MEV vulnerability analysis across seven DAG-based consensus protocols, implementing three novel frontrunning attack strategies: Fissure, Speculative, and Sluggish, to measure Attack Success Rates (ASR). Our findings demonstrate that protocol architecture fundamentally determines MEV resistance, with two-layer architectures showing significantly higher resilience than single-layer designs.

Background: From BFT to DAGs

Practical Byzantine Fault Tolerance (PBFT) has long served as the gold standard for BFT protocols, utilizing a leader-based multi-phase communication approach to reach consensus. However, PBFT faces significant hurdles: **complex view-change mechanisms**, **heavy leader workloads**, and the **entanglement of message dissemination** with total ordering.



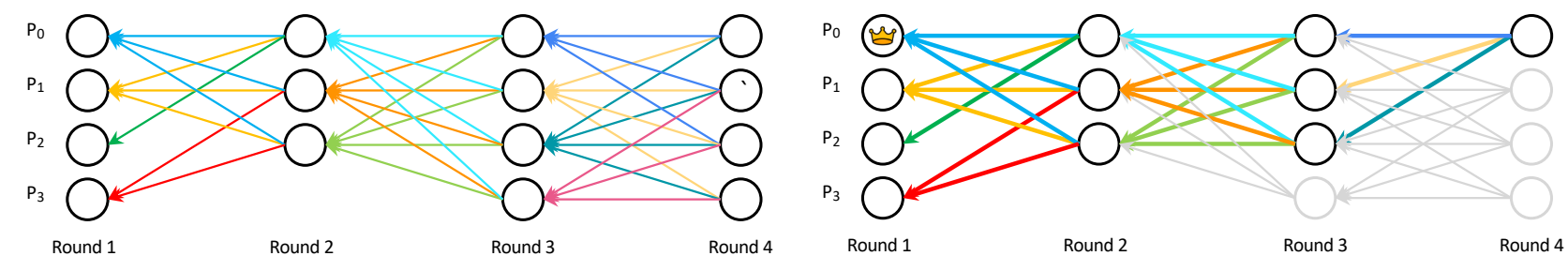
challenge 1: Complex View-Change Mechanism

challenge 2: Leader Bottleneck & Inefficient Resource Utilization

challenge 3: Entanglement of Dissemination and Ordering

The Rise of DAG-based BFT

While traditional BFT protocols like HotStuff eliminate separate view-change routines through synchronized communication phases, they often fail to optimize resource utilization across all nodes. Parallel ordering protocols such as MirBFT and RCC attempt to address this by running multiple simultaneous BFT instances, yet they still require complex mechanisms to establish a total ordering among replicas. Other approaches like DispersedSimplex utilize erasure-coded information-dispersal schemes to harness network bandwidth, but these introduce significant computational and synchronization overhead. DAG-based protocols resolve these collective shortcomings by decoupling dissemination from consensus, enabling high-throughput parallel processing without the bottlenecks of leader-centric or complex dispersal designs.



DAG-based protocols represents a fundamental decoupling of **data dissemination** from **consensus** logic. In this paradigm, nodes continuously broadcast blocks to build a local Directed Acyclic Graph, effectively utilizing the cumulative bandwidth of all network participants [2].

Core Mechanisms of DAG-based Protocols

- Parallel Dissemination:** Nodes continuously broadcast blocks to build a local DAG, utilizing the bandwidth of all parties [3].
- Zero-Message Ordering:** Consensus is achieved by "interpreting" the DAG locally, eliminating the need for separate communication for every transaction [3].
- Modern Adoption:** This architecture is the foundation for high-performance systems like Sui (Bullshark/Mysticeti) and Aleph Zero (AlephBFT).

From Performance to Vulnerability

While DAG-based protocols provide massive scalability, they introduce new degrees of freedom in the transaction ordering process. The decentralized nature of DAG building—where nodes choose which "parent" blocks to reference—creates an expanded attack surface for Maximal Extractable Value (MEV) reordering strategies.

MEV: Definition and Categorization

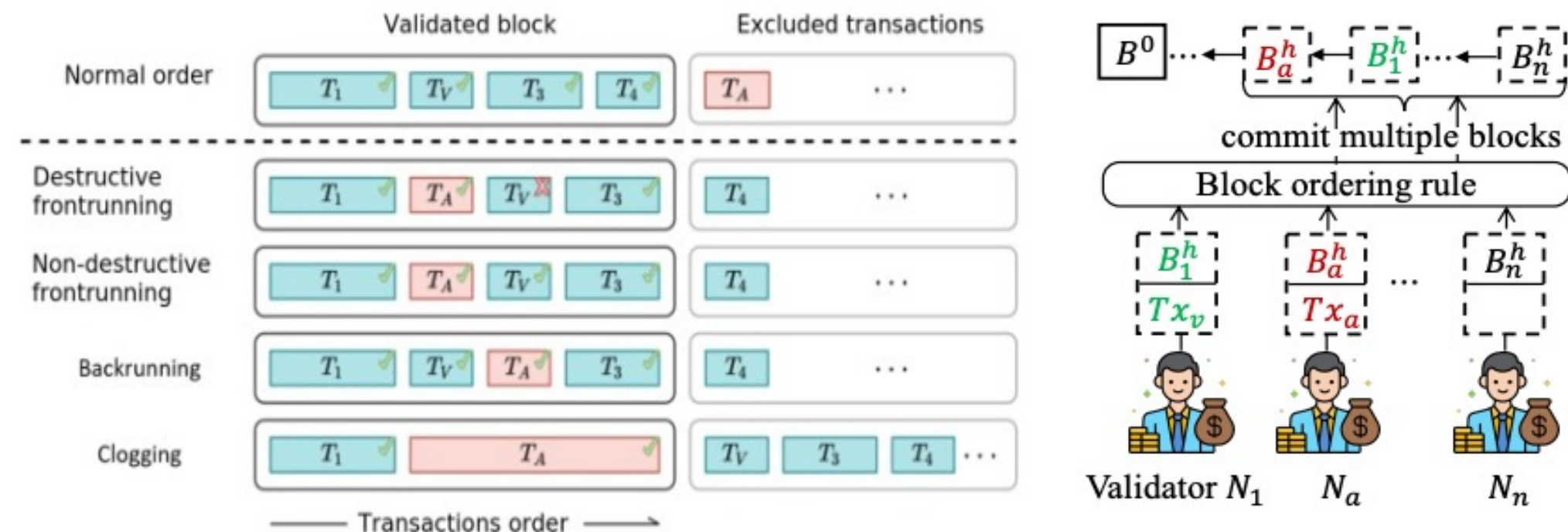
Maximal Extractable Value (MEV) refers to the total amount of economic value that block producers (validators, miners, or sequencers) can extract by arbitrarily reordering, including, or excluding transactions during the block production process. Originally termed "Miner Extractable Value," the concept has evolved as a systemic challenge for all smart-contract blockchains where transaction sequencing is discretionary.

Common MEV Strategies

Frontrunning: A malicious actor identifies a profitable transaction in the mempool and inserts their own identical transaction with a higher fee to ensure it is processed first.

Sandwich Attacks: An attacker wraps a victim's large trade between a buy order (frontrun) and a sell order (backrun) to profit from the price slippage caused by the victim's transaction.

Arbitrage and Liquidations: Searchers use automated bots to exploit price discrepancies across decentralized exchanges or to be the first to liquidate under-collateralized loans.



Sandwich Attack Example: the "Block N" execution sequence, the attacker first executes a **front-run** (buy) transaction to artificially inflate the asset's price just before the user's swap occurs. After the victim is forced to trade at this disadvantageous price, the attacker immediately executes a **back-run** (sell) transaction to offload their assets and capture the price slippage as guaranteed profit.

Categorization of Attacks MEV attacks can be broadly categorized based on their impact and mechanism:

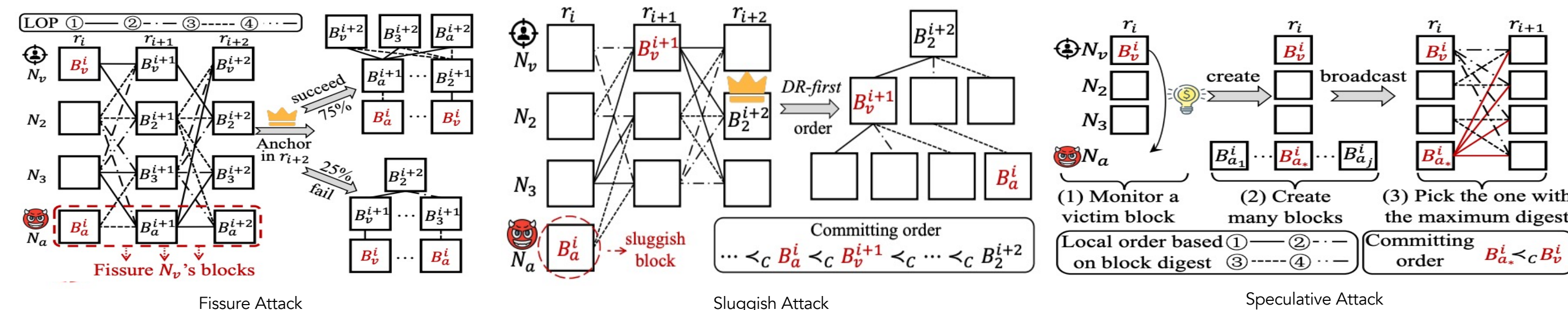
- Market-Based (L2):** Strategies like arbitrage and sandwiching that exploit application-layer logic and user slippage settings.
- Consensus-Layer (L1):** Sophisticated attacks such as **Time-Bandit** or **Uncle-Bandit** attacks, where validators reorganize block history or selectively "resurrect" transactions from rejected blocks to capture value.
- Network-Specific (DAG):** As explored in this research, protocols utilizing Directed Acyclic Graphs introduce new vectors—**Fissure**, **Sluggish**, and **Speculative** attacks—that exploit the unique parent-selection and round-ordering rules of high-performance consensus.

MEV Attacks on DAGs

While traditional MEV exploits application-layer logic (e.g., sandwich attacks on DEXs), DAG-based systems are uniquely susceptible to consensus-layer manipulation. Attackers exploit the decentralized **parent-selection** and **round-ordering** rules to resequencing transactions for profit [1].

Attack Taxonomy

- Fissure Attack:** An attacker strategically excludes specific parent links in the DAG to delay a victim's block. By withholding these references, the attacker prevents the victim's block from achieving early certification, ensuring the attacker's own transactions are ordered first in the final ledger [1].
- Sluggish Attack:** This strategy exploits the timing-dependent nature of round-based ordering. By injecting intentional processing delays into certificate broadcasts, an attacker can manipulate the perceived time or round of a transaction, causing it to be ordered after the attacker's own blocks [1].
- Speculative Attack:** Targeted at protocols using lexicographical tie-breaking rules, attackers utilize "digest grinding"—mining thousands of block hashes—to find a specific hash that wins the tie-breaking competition, granting them ordering priority in "tie" scenarios within the same round [1].



Experimental Methodology, To measure protocol vulnerabilities, we implemented and run these attacks on 7 different protocols

- Environment:** A 13-node network with 4 attackers and 3 victims. ($f = 4$)
- Primary Metric:** Attack Success Rate (ASR).
- Benchmark:** An ASR of approximately 50% represents a fair, random ordering (baseline), while values significantly higher indicate successful exploitation.

$$ASR = \frac{\text{success}}{\text{total pairs}}$$

Results: Protocol Vulnerability Ranking

Our analysis of 7 protocols reveals that **architecture** is the primary driver of security:

Protocol	Baseline	Fissure	Speculative	Sluggish	Best Attack
Narwhal-Tusk	50%	88.5%	70.1%	99.9%	Sluggish (99.9%)
Bullshark	50%	91.9%	97.8%	89.9%	Speculative (97.8%)
MEVSUI	50%	90.9%	92.5%	92.2%	Speculative (92.5%)
Mahi-Mahi	50%	86.0%	52.0%	44.0%	Fissure (86.0%)
AlephBFT	31%	52.3%	0.0%	54.1%	Sluggish (54.1%)
Mysticeti	47%	45.0%	40.0%	40.0%	None (~47%)
Autobahn	50%	~50%	N/A	~50%	None (~50%)

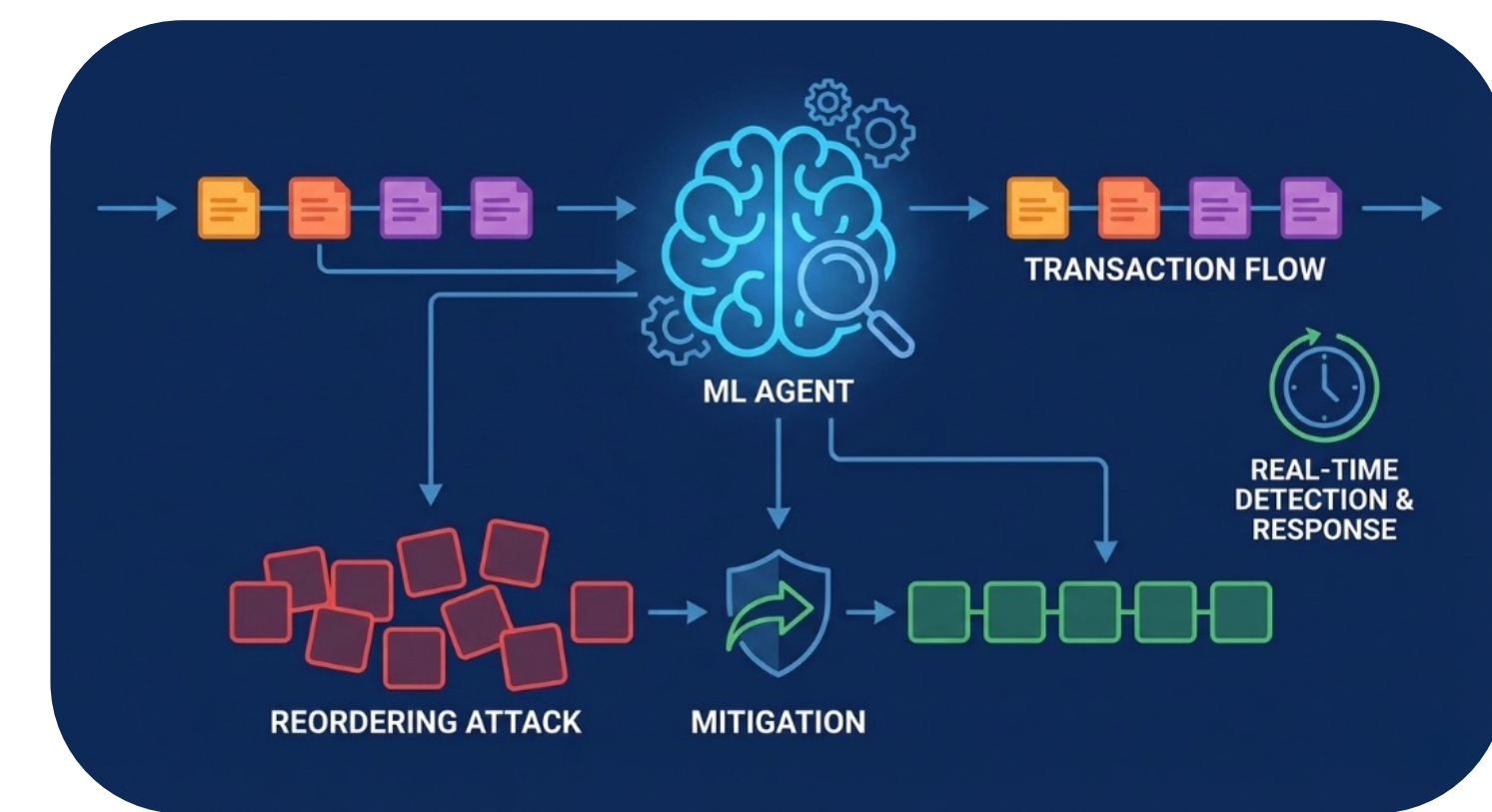
Key Insights & Architectural Defenses

- Consensus-Layer Aggregation:** Protocols like Autobahn neutralize attacks by collecting proposals from all nodes, preventing single-node exclusions.
- Two-Layer Isolation:** Separating the mempool (proposal) from the consensus layer provides a critical security barrier.
- Virtual Voting:** Used by AlephBFT, this dilutes attacker influence by aggregating across the entire network.

Architectural Defenses and Conclusions

The analysis reveals that Autobahn is the most resistant protocol due to its two-layer architecture and consensus-layer aggregation, which **isolates** the mempool from consensus decisions. In contrast, single-layer protocols like Bullshark and its forks remain highly vulnerable despite MEV mitigation patches. To ensure MEV resistance, protocol designers should prioritize round-only ordering and require global certification before commitment. Future work will focus on developing Reinforcement Learning (RL) agents to autonomously detect and mitigate these attack vectors in real-time.

Future Directions



- Developing **ML-driven agents** to autonomously detect and mitigate sophisticated reordering attacks in real-time.
- Analyzing the **economic impact** of DAG-specific MEV in geo-distributed network deployments.

References

- Zhang, J. and Kate, A., 2024. No fish is too big for flash boys! frontrunning on DAG-based blockchains. Cryptology ePrint Archive
- Nagda, H., Sankhe, S., Sinha, S., Attarha, K., Amiri, M.J. and Loo, B.T., 2025. DAG of DAGs: Order-Fairness Made Practical. Proceedings of the ACM on Management of Data, 3(6), pp.1-27.
- Keidar, I., Kokoris-Kogias, E., Naor, O. and Spiegelman, A., 2021, July. All you need is dag. In Proceedings of the 2021 ACM Symposium on Principles of Distributed Computing (pp. 165-175).